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[4757]-1074

S.E. (Computer Engg.) (First Semester) EXAMINATION, 2015

OPERATING SYSTEM AND ADMINISTRATION

(2012 PATTERN)

Time : Two Hours

Maximum Marks : 50

- N.B. :—** (i) Attempt Q. No. 1 or Q. No. 2, Q. No. 3 or Q. No. 4,
Q. No. 5 or Q. No. 6, Q. No. 7 or Q. No. 8.
- (ii) Neat diagrams must be drawn wherever necessary.
- (iii) Figures to the right indicate full marks.
- (iv) Assume suitable data, if necessary.

1. (a) Draw and explain high level architecture of unix systems. [6]
- (b) Write a short note on interrupts and exceptions. [4]
- (c) Explain batch processing and multiprogramming. [2]

Or

2. (a) Explain 4 state process diagram. [6]
- (b) What do you mean by user and hardware perspective of operating system ? [4]
- (c) List any two functions of an OS. [2]

P.T.O.

3. (a) Explain organization of the file hierarchy tree. [6]
(b) Explain types of files in unix. [3]
(c) Explain input, output and error redirection. [3]

Or

4. (a) Explain the entire process of booting (system turn on to login prompt). [6]
(b) Differentiate between grub and LILO. [4]
(c) Define a shell. [2]
5. (a) Which *two* signals can never be ignored ? [2]
(b) Explain signal and kill function with parameters. [4]
(c) What are real effective and save ids. [3]
(d) Differentiate between nice and renice (with examples). [4]

Or

6. (a) Differentiate between user mode and kernel mode instructions. [4]
(b) How does a user level process change to kernel mode ? Explain. [2]
(c) Explain how kill function can be used to send signal to a group of processes also give the syntax of kill function. [4]
(d) What is the shadow password file ? Where does user password reside in unix. [3]

7. (a) Write a short note on RAID and its levels. [6]
(b) Explain the process of mounting and unmounting a pendrive. [4]
(c) Explain the fields in /etc/group file. [3]

Or

8. (a) What is meant by disk partitioning and disk formatting. [4]
(b) What is the native file of Linux and Unix. Differentiate between Unix and Linux. [4]
(c) What are the startup files ? What are its tasks ? [3]
(d) How to set the password for user account in linux/unix. [2]